GRC Transactions, Vol. 41, 2017

Comparison and Controls of Thermal Spring Chemistry in Cascade Range and Olympic Mountains Geothermal Provinces, Washington

Jon K. Golla and Jeffrey H. Tepper

Geology Department, University of Puget Sound, Tacoma WA

Keywords

geochemistry, Cascades, Olympics, geology, rock-water, geothermometry, stable isotopes, conceptual models

ABSTRACT

Early geothermal surveys conducted by the Washington State Department of Natural Resources - Geology and Earth Resources Division from the late 1970's to the early 1980's identified significant hydrogeochemical variation among Washington thermal springs. This study characterizes five thermal springs in the Cascade Range (Carson, Bonneville, Baker, Ohanapecosh, and Goldmyer) and two in the Olympic Mountains (Olympic Complex and Sol Duc) using geochemical indicators (solute chemistry, isotopic signatures, and geothermometry estimates) commonly employed during geothermal exploration. We conclude that the variations result from differences in rock-water interactions (dissolution and low-grade alteration) as well as features of the systems, including source water, heat source, reservoir connectivity, and topographic relief.

The Cascade thermal springs occur in a geothermal environment that is influenced by magmatic fluids, as supported by the widespread volcanic activity in the region. Chemical enrichment ($\bar{x}_{TDS}$: 1198 mg/L) and diversity (i.e., Cl: 94-683 mg/L) may be results of magmatic fluid contribution, topographic lateral flow, and exposure to a diverse suite of igneous rocks. Inherently, Cascade geothermal fluids are also hotter (53-153 ºC) and seem to undertake a more indirect path to the surface, as evidenced by signs of dilution (Carson = 24%) and degassing (Bonneville = 23% and Ohanapecosh = 2%) and significant differences ($\bar{x}_{difference}$: 53 ºC) between discharge and estimated reservoir temperatures. Conversely, the Olympic thermal springs are manifestations of a cyclic geothermal system that is controlled by subvertical thrust faults. These chemically dilute ($\bar{x}_{TDS}$: 355 mg/L) and homogeneous (i.e., Cl: 6-12 mg/L) spring waters are probably salinized by salts in or connate waters of the turbidite host rocks. The cooler (53-103 ºC) Olympic geothermal fluids appear to be more directly connected to their reservoir, as differences ($\bar{x}_{difference}$: 27 ºC) between discharge and subsurface temperatures are smaller. Oxygen (-15.12‰ to -10.52‰) and hydrogen (-116.18‰ to -73.55‰) stable isotopic signatures suggest that both geothermal provinces are likely recharged by local meteoric waters.

1. Introduction

The state of Washington hosts many low-temperature geothermal resources. Most are in western Washington, with many widely spread out along the Cascade Range and some tightly clustered in the northern Olympic Mountains. Some eastern regions (e.g. Yakima Fold Belt, Palouse

Slope) have an elevated geothermal gradient, but these areas have been less extensively documented (Blackwell et al., 1990; Schuster and Bloomquist, 1994). During the late 1970's and early 1980's, in response to the prevalent energy crises and with the funding of the U.S. Department of Energy, the Washington State Department of Natural Resources - Geological and Earth Resources Division investigated possible commercialization of local geothermal prospects (Bloomquist and Schuster, 1994; Korosec et al., 1983). Early reports identified resources suitable for low temperature or direct use, but progress was eventually barred by recovery of the oil and energy market and by water rights issues (Bloomquist and Schuster, 1994). In recent years, there has been renewed interest in Washington's geothermal resources, particularly in the Wind River Valley, Mt. St. Helens, and Mt. Baker vicinities (Boschmann et al., 2014; Czajkowski et al., 2012; Forson et al., 2016).

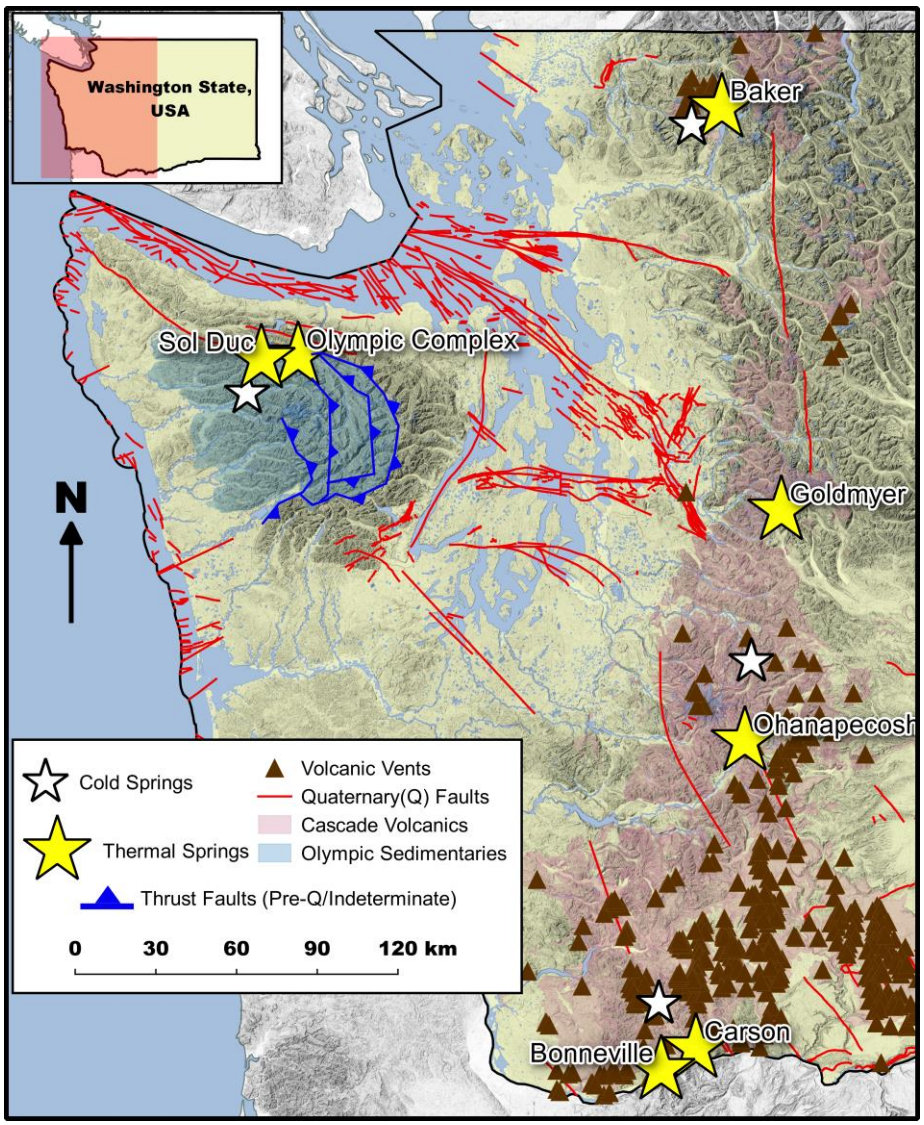


Figure 1: Map of the study area indicating geochemical collection sites and showing simplified geologic and tectonic differences between Cascade Range (back arc) and Olympic Mountains (forearc). Areas of higher elevation are shaded darker (in green).

This study focuses on seven thermal spring sites: five in the Cascades (Goldmyer, Carson, Baker, Bonneville, and Ohanapecosh) and two in Olympics (Olympic Complex, and Sol Duc), but also includes four neighboring cold springs for chemical comparisons (Figure 1). Previous geothermal surveys recognized significant hydrogeochemical variation among these thermal springs. Our research aims to better understand the origins of this diversity by using geochemical techniques commonly applied in geothermal exploration to assess the importance of differences in (1) source waters, (2) subsurface temperatures, (3) reservoir lithology, and (4) features of the housing geothermal systems.

2. Geologic Context

The bedrock geology of central and western Washington reflects its convergent margin setting. The Olympic Peninsula contains oceanic basalts that originated as part of the Siletzia terrane, a large igneous province that was accreted to North America about 50 My ago (Wells et al., 2014). The material thrusted beneath these basalts are marine sediments (turbidites) scraped off the downgoing Juan de Fuca plate during subsequent subduction. Beginning around 40 Ma, this subduction also led to the onset of arc magmatism and establishment of the modern Cascade arc (Christiansen and Yates, 1992).

The geology of the Wind River Valley, which hosts Carson and Bonneville springs, consists of Oligocene subaerial basalts and volcaniclastics (Wise, 1961 and 1970; Boschmann et al., 2014). The area of Baker spring is underlain by the Chilliwack Batholith, which consists of plutonic rocks ranging from gabbro to granite (Tabor et al., 2003). Goldmyer spring issues from the Miocene Snoqualmie batholith (Korosec et al., 1983). Geology in the Ohanapecosh spring site comprises of coarse volcaniclastics and dominantly andesitic flows of the Ohanapecosh Formation (Fiske et al., 1963; Frank, 1995).

Conversely, the geology of the springs in the Olympic Mountains contains thrust-imbricated marine sedimentary rocks from the Cascadia accretionary wedge material and oceanic crust (Brandon and Vance, 1992; Tabor and Cady, 1978). The general region consists of marine turbidites with interbedded carbonate and some metasedimentary rocks, such as slates and schists (Tabor and Cady, 1978).

Most springs (Baker, Ohanapecosh, Olympic Complex, Goldmyer, and Sol Duc) are in areas of relatively high elevation with moderate-to-high relief (Figure 1; Table 1). Two springs (Carson and Bonneville) are in a low-elevation setting with gentle topography (Figure 1; Table 1). Tectonically, the Cascade springs are associated with local volcanism, while Olympic springs are seated to Pre-Quaternary (or indeterminate in age), eastward dipping thrust faults that are now subvertical from tectonic and erosional exhumation (Figure 1; Brandon et al., 1998).

Table 1: Field data and water chemistry of sampled springs. All ionic concentrations are reported in mg/L, unless stated otherwise.

Parameter	Carson	Bonneville	Baker	Ohanapecosh	Goldmyer	Olympic 1	Olympic 2	Olympic 3	Olympic 4	Sol Duc	Silver	Tyee	Schreiber's	Taft
Lat (N)	45.65595	45.72786	48.76125	46.73747	47.48371	47.97657	47.97639	47.97639	47.97615	47.968719	46.99584	45.875749	48.707364	47.860766
Long (W)	121.96065	121.79536	121.67179	121.56263	121.38901	123.68939	123.68831	123.68771	123.68639	123.86250	121.53167	121.97185	121.81609	123.93249
Elev.	21	79	449	578	574	614	618	616	610	504	816	343	1070	204
Date	6/3/2016	6/3/2016	6/10/2016	6/11/2016	6/15/2016	6/13/2016	6/13/2016	6/13/2016	6/13/2016	6/21/2016	6/28/2016	7/2/2016	7/8/2016	7/9/2016
pH	8.03	9.39	8.48	7.22	8.97	9.71	9.79	9.59	9.53	9.53	6.79	6.85	6.62	6.77
T (°C)	36.0	45.0	42.4	37.4	44.0	45.9	43.9	43.8	39.2	49.7	7.0	8.3	5.6	8.7
DO	1.20	0.80	0.21	0.21	2.60	1.00	2.60	0.75	3.81	1.60	9.40	10.77	11.98	9.60
TDS	1194	555	639	3161	440	360	363	314	317	375	79	-	164	68
HS (μg/L)	978	90	1290	0	111	4123	4610	3995	3715	3710	7	19	10	6
Na	326	133	153	859	117	64.9	64.5	63.7	63.3	75.2	3.3	2.8	12.9	2.1
K	5.22	0.67	8.68	46.47	2.88	0.82	0.80	0.74	0.83	0.34	-	-	4.09	-
Ca	70.9	38.6	5.4	53.6	6.3	1.0	1.0	2.1	1.2	1.0	8.2	5.1	8.8	11.3
Mg	1.1	0.2	0.3	4.3	0.1	0.1	0.1	0.3	0.1	0.1	1.0	1.0	6.2	1.3
SiO_2	62.3	58.6	129	123	72.3	78.4	77.8	75.2	78.8	76.5	28.1	21.2	56.2	8.9
Cl	683	189	94.0	906	129	6.8	6.6	6.9	7.7	12.6	1.5	0.9	4.5	1.3
F	1.3	1.4	3.7	7.5	1.7	1.1	1.3	0.8	1.0	1.7	0.2	0	0.5	0
SO_4	18.5	84.8	90.4	179	44.9	20.8	19.8	17.5	13.3	17.1	8.7	5.1	55.9	10.9
HCO_3	19.1	34.7	144	997	59.5	159	161	124	128	168	29.9	24.6	16.6	34.8
Li	2.49	0.04	0.34	2.27	0.13	0.04	0.04	0.03	0.03	0.05	-	-	0.01	-
B	2.49	2.58	2.65	9.37	0.66	0.78	0.83	0.78	0.88	1.20	-	-	0.17	-
Sr	0.07	0.03	0.23	1.54	0.09	0.07	0.07	0.07	0.07	0.05	-	-	0.06	-
Ba	0	0.01	0.02	0.09	0	0.01	0.01	0.01	0	0	-	-	0	-
Fe	0	0	0.13	0.37	0.01	0.02	0.01	0	0.06	0.02	0.01	0.06	0	0
Mn	0.08	0.07	0.09	0.06	0	0.08	0.08	0.08	0.07	0.08	0.07	0.07	0.07	0.07
Al	0.01	0.01	0.36	0.01	0.02	0.06	0.04	0.03	0.11	0.06	0	0.62	0.01	0
δ18O	-11.53	-11.23	-13.76	-15.12	-13.34	-13.46	-13.48	-12.95	-13.45	-12.56	-14.35	-10.94	-14.63	-10.52
δD	-88.59	-77.42	-100.72	-116.18	-95.07	-96.76	-97.08	-93.6	-97.06	-89.23	-104.69	-76.05	-105.24	-73.55
CBE	-5%	0%	1%	-5%	-1%	-6%	7%	5%	4%	-3%	-7%	-7%	1%	-5%

(-) not determined

(Elev.) Elevation in meters

* measured in μg/L

3. Sampling and Analytical Techniques

Sixteen sets of water samples were collected from seven thermal (Carson, Bonneville, Ohanapecosh, Baker, Olympic, Sol Duc, Goldmyer) and four cold (Silver, Tyee, Taft, Schreiber's Meadow) springs (Figure 1). All springs were easily accessible, and most discharged to commercial or natural pools. A polyethylene bottle was submerged beneath the surface with the mouth facing slightly downwards, and then filled and capped tightly underwater to obtain samples directly. Where numerous seeps were present in one site, multiple springs were sampled. At each spring, three bottles were filled for analysis of: (1) anions and alkalinity, (2) major and trace cations and (3) H and O isotope analyses. Samples for ion analyses were filtered through a 0.45μm membrane. Time-sensitive parameters (pH, dissolved oxygen (DO), temperature, and H_2S) were measured with a YSI multiparameter probe and a Hach DR-2400 portable spectrophotometer (Table 1).

Total alkalinity (expressed as HCO_3) was measured within 24 hours of sampling via Gran titrations (Wetzel and Likens, 2000). Major cations (Na, K, Ca, Mg, Si) were measured at the University of Puget Sound on a Perkin Elmer DV 2000 Inductively Coupled Plasma - Optical Emission Spectrometer (ICP-OES); sample bottles used for this analysis were acidified with 2% HNO3 after filtration. Major anion (Cl, F, SO_4, I) concentrations were obtained with a Dionex ICS-4000 Integrated Capillary Ion Chromatograph (IC) in the Department of Geosciences at Pacific Lutheran University. Trace (Li, B, Sr, Ba, Fe, Mn, and Al) element concentrations were determined by Inductively Coupled Plasma - Mass Spectrometer (ICP-MS) in the University of Washington - Tacoma. Spring waters were also analyzed for stable water isotopes (δD and δ18O) via Picarro Cavity Ring-Down Spectroscopy (CRDS) method at IsoLab of University of Washington - Seattle. Sample bottles for this analysis were collected with minimal headspace to minimize any evaporation or any exchange with atmospheric water vapor. The recorded δD and $\delta^{18}O$ values are reported in standard per mil (‰) notation relative to Vienna Standard Mean Ocean Water (VSMOW) international scale via VSMOW and Standard Light Antarctic Precipitation (SLAP). Charge balance errors (CBE) and total dissolved solids (TDS) are also reported (Table 1). Most samples lie within the acceptable range of ±5% CBE; four samples (Olympic 1, Olympic 2, Silver, Tyee) have CBE a few percent (1-3%) outside the acceptable window, which may be due to minor instrumental error or other sampling malpractices.

4. Water Chemistry

4.1 Chemical Composition

The aqueous chemistry data are summarized in Table 1 and plotted in Figures 2,3, and 4 using Powell and Cumming's (2010) spreadsheets. Generally, thermal spring discharges are ≤ 50 ºC, although the springs in the Olympics (43.8-49.7 ºC) are generally warmer than those at the Cascades (36.0-45.0 ºC). All thermal springs are slightly-to-moderately alkaline, as Cascade and Olympic waters respectively show pH ranges of 7.22-9.39 and 9.53-9.79. Hydrogen sulfide (expressed as dissolved HS^-) is generally low in Cascade springs (0-1290 μg/L), but more abundant in Olympic springs (3710-4610 μg/L). Conversely, trace elements in Cascade springs are significantly more concentrated than in Olympic springs. This is particularly evident for boron ($\bar{x}$: 3.55 mg/L vs 0.89 mg/L) and lithium ($\bar{x}$: 1.05 mg/L vs 0.04 mg/L).

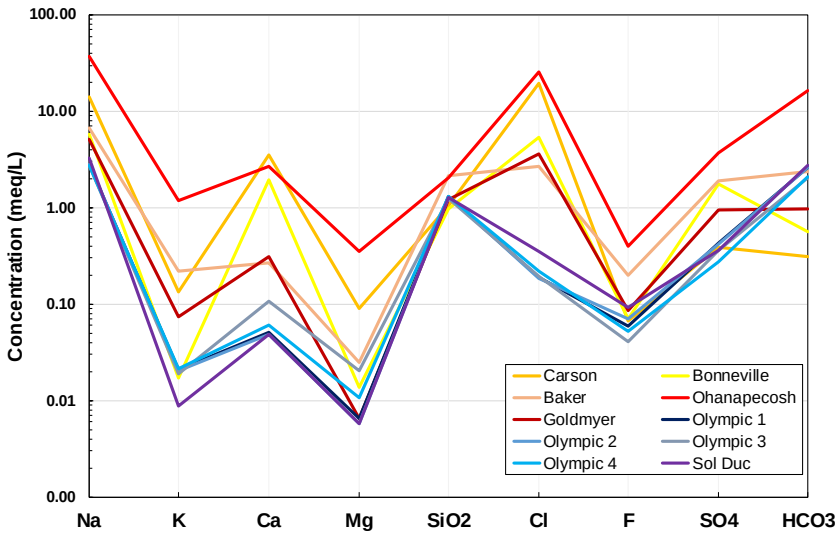


Figure 2: Schoeller diagram showing concentrations of major ions. Warm colors represent Cascade thermal waters, while cool colors symbolize Olympic thermal waters.

Cascade thermal waters are significantly more diverse and more enriched in dissolved major constituents compared to the springs in the Olympics (Figure 2). The Cascade thermal springs (Na- and Cl-dominant) are also noticeably different from cold springs (Ca-/Mg- and HCO_3-rich) (Figure 3a). Conversely, Olympic thermal springs differ from nearby cold springs only in that they are more Na-rich (Figure 3a). Based on major anion abundances, the thermal springs can be divided into three groups: mature waters (Carson), intermediate waters (Bonneville, Baker, Ohanapecosh, Goldmyer), and peripheral waters (Olympic complex, Sol Duc) (Figure 3b). Thermal springs in this study have very similar compositions to those reported in Korosec et al. (1983) and Mariner et al. (1982) and fit within the array of published analyses of Washington thermal and mineral springs in Figure 3 (Mariner et al., 2006).

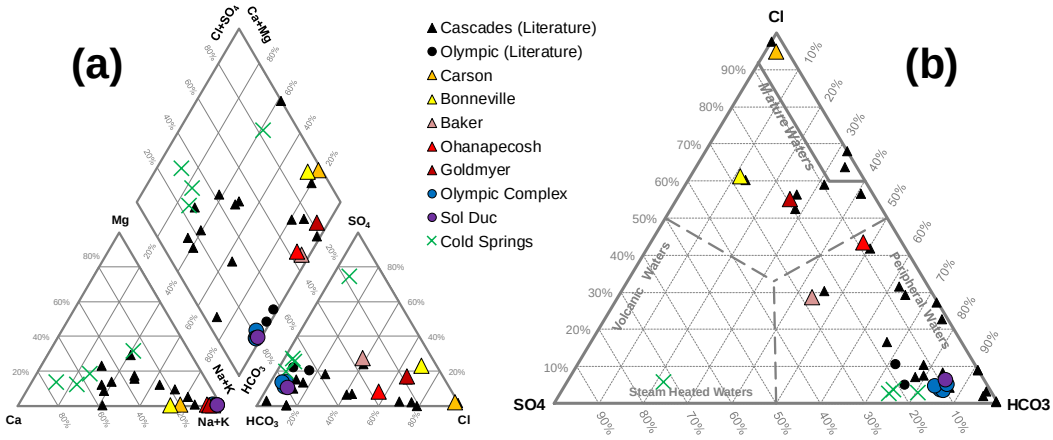


Figure 3: Chemical plots of thermal and cold springs data. Piper plot (a) shows abundances of major ions relative to each other. Anion ternary diagram (b) further classifies waters into distinct, geothermal natures.

4.2 Isotopic Signatures

Oxygen and hydrogen stable isotopic compositions ($\delta^{18}O$ and δD) of Cascade and Olympic springs range from -15.12‰ to -10.52‰ and -116.18‰ to -73.55‰ respectively (Table 1). These signatures, along with literature data from Mariner et al. (2006), are plotted in Figure 4 together with the Global Water Meteoric Line (GWML) and the Pacific Northwestern Local Meteoric Line (PNW-LMWL) derived by Craig (1961) and Kendall and Coplen (2001).

Except for a few springs (Carson, Ohanapecosh, Goose Egg Soda Spring, and Alpha Mineral Spring), most waters lie along the meteoric water lines, suggesting that western Washington springs are dominantly recharged by meteoric waters (Figure 4). Carson and Ohanapecosh thermal springs show small departures towards heavy isotopic enrichment, while Alpha Mineral Spring and Goose Egg Soda Spring deviate more significantly. These shifts can potentially be explained by rock-water interactions, regional evaporation, and/or mixing with andesitic water derived from subducted slab (Giggenbach, 1992). In the rock-water scenario, $\delta^{18}O$ shifts positively when ascending spring water exchanges with heavier oxygen isotopes of accessible reservoirs (Taylor, 1974; White et al., 1973). A local evaporation trend and mixing with magmatic water would produce positive shifts in both $\delta^{18}O$ and δD (Craig et al., 1963; Giggenbach, 1992). However, evaluating either of these scenarios will require more isotopic data. Considering that the Carson and Ohanapecosh departures from the meteoric water lines are slight, they are probably meteoric in origin, too.

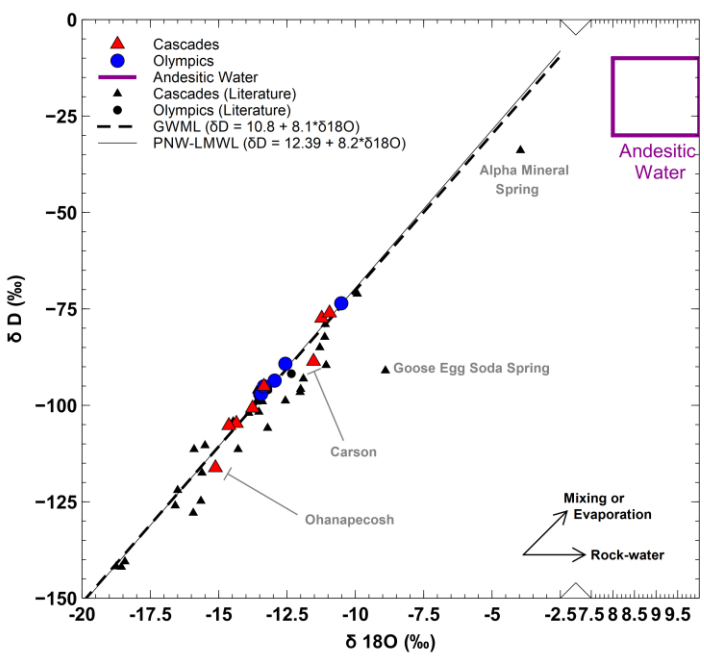


Figure 4: Water stable isotope ratios of thermal and cold springs compared to literature data of Washington springs, GWML (Craig, 1961), PNW-LMWL (Kendall and Coplen, 2001), and worldwide average of convergent margin andesitic waters (Giggenbach, 1992). Vienna Standard Mean Ocean Water (VSMOW) and Standard Light Antarctic Precipitation (SLAP) were used as reference standards. Labeled springs are used to highlight deviations from water lines. Black arrows indicate physical and kinetic trends that may be responsible for these shifts.

5. Geothermometry

Conventional silica and cation geothermometers are used to determine reservoir temperatures for these springs, although their limitations should be considered (Table 2). Temperatures provided by silica geothermometers are affected by dilution or mixing (Fournier, 1977). The Na-K geothermometers require equilibrium between the geothermal fluid and feldspars, such as albite and K-feldspar (Fournier, 1979). The Na-K-Ca geothermometer assumes equilibrium between geothermal waters and feldspars and calcite or Ca-bearing minerals, so any addition of bicarbonate waters that causes calcium to decrease through precipitation will lead to inaccurate estimates (Fournier and Truesdell, 1973). The limited scope of the present study makes it difficult to account for any of these complications. Furthermore, the chemical homogeneity of Olympic waters and wide geographic spread of the Cascade springs (which do not belong to the same geothermal system) precludes construction of classical mixing models (i.e., silica-enthalpy, chloride-enthalpy, Giggenbach ternary) (Fournier, 1977; Giggenbach, 1988).

An alternative method of determining reservoir temperatures is Multiple Mineral Equilibration (MME) geothermometry (Reed and Spycher, 1984). This method uses the full chemical analysis of spring waters to compute the degree of saturation (log(Q/K)) of likely reservoir minerals over a range of temperatures (25-225 ^{o}C). The degrees of saturation are plotted as a function of temperature (Figure 5). Assuming that the geothermal water was equilibrated with the reservoir host rocks, the point at which the mineral curves cluster at log(Q/K) = 0 equilibrium yields the reservoir temperature. Note that the chemical equilibrium MME estimates refer to may be local and not indicative of the true geothermal reservoir. All speciation calculations were done through SOLVEQ (Reed et al., 2016).

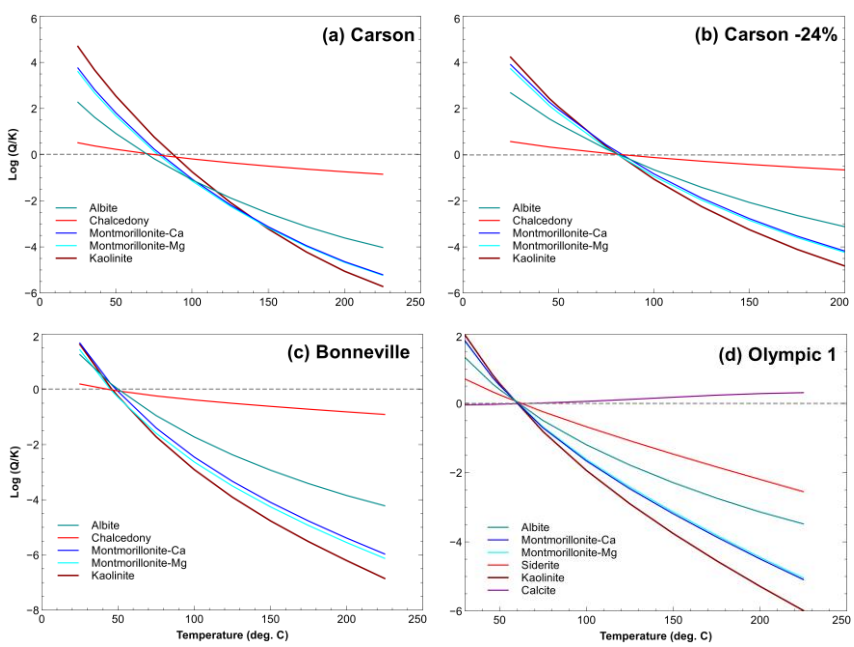


Figure 5: MME plots for a) Carson sample showing effects of dilution, b) Carson sample corrected by taking out GW component, c) Bonneville sample showing slight effects of CO_2 degassing, and d) Olympic 1 sample requiring no corrections. Diagenetic ± authigenic minerals considered for reservoir assemblage.

A good MME estimate is indicated by tight convergence of equilibrium lines at log(Q/K)=0 as seen in some samples (e.g., Olympic 1, Figure 5d; Table 2). On the other hand, convergence off this line or loose clustering of curves suggests the water has changed composition since leaving the reservoir or that equilibrium was not attained. One Carson water sample shows convergence within undersaturation (log(Q/K) < 0), which implies dilution or mixing (Figure 5a). For diluted or mixed waters, a mass balance correction can be made:

$$Sample = (Geothermal * x) + \left(GW * (1 - x)\right) \quad (1)$$

Equation 1 assumes that a sample of spring water consists of a mixture of geothermal water and shallow, dilute groundwater (GW). Reconstructing the Carson geothermal component is done by taking out 24% of USGS Well, a nearby GW analog obtained from the USGS National Water Information System database, and results in a tighter clustering of curves (Figure 5b).

Conversely, a Bonneville water sample shows clustering above the equilibrium line (log(Q/K)>0), which suggests degassing of CO_2 (Figure 5c; Pang and Reed, 1998). The CO_2 add-back correction proposed by Pang and Reed (1998) is applied. The Bonneville water sample is corrected by adding 23% more CO_2 to account for any changes in pH that are caused by escape of gas.

Table 2: Results of geothermometry calculations in ºC, including adjustments made for MME estimates for all thermal spring samples. Discharge temperatures are also reported for comparisons.

Spring	Discharge (ºC)	Alpha Cristobalite	Chalcedony	Quartz (conductive)	Quartz (adiabatic)	Na-K-Ca	Na-K-Ca (Mg-corrected)	Na/K (Giggenbach, 1988)
Carson	36.0	62	83	113	112	63	63	119
Bonneville	45.0	59	80	110	109	14	14	70
Baker	42.4	102	127	152	145	161	153	191
Ohanapecosh	37.4	99	124	149	143	168	140	188
Goldmyer	44.0	69	91	120	118	85	85	141
Olympic 1	45.9	73	96	124	122	77	77	108
Olympic 2	43.9	73	96	124	121	77	77	107
Olympic 3	43.8	71	94	122	120	58	58	104
Olympic 4	39.2	74	96	125	122	73	73	109
Sol Duc	49.7	72	95	123	121	53	53	66

Spring	Discharge (ºC)	MME Estimate	Position of Convergence	Correction [quantity]	Final MME Estimate
Carson	36.0	68 - 88	Undersaturation	Dilution [24%]	81
Bonneville	45.0	43 - 54	Oversaturation	Degassing [23%]	53 - 57
Baker	42.4	153	Equilibrium	N/A	153
Ohanapecosh	37.4	98 - 107	Oversaturation	Degassing [2%]	108
Goldmyer	44.0	72	Equilibrium	N/A	72
Olympic 1	45.9	61	Equilibrium	N/A	61
Olympic 2	43.9	52	Equilibrium	N/A	52
Olympic 3	43.8	53-56	Equilibrium	N/A	53 - 56
Olympic 4	39.2	84	Equilibrium	N/A	84
Sol Duc	49.7	103	Equilibrium	N/A	103

Table 2 shows reservoir temperature estimates and measured discharge temperatures for all thermal springs. Excluding likely erroneous Na-K-Ca estimates for Bonneville, most Cascade waters are predicted to be hotter at depth than Olympic waters. The average difference between discharge and subsurface temperatures is greater in the Cascade springs ($\bar{x}_{difference}$: 53 ºC vs 27 ºC). In addition, these waters show more evidence of chemical evolution since ascent, as evidenced by corrections for dilution and degassing (Table 2). These observations suggest that Cascade thermal waters have taken a slower/more indirect path to the surface.

6. Chemical Controls

Changes in water chemistry in geothermal systems are mainly governed by interactions with minerals in subsurface and thermal waters, magmatic fluid inputs, and degassing. In addition, some observed differences are probably caused by differences in the physical and structural characteristics of the system.

6.1 Rock-Water Interactions

Higher levels of Cl (94.0-906 mg/L), SO_4 (18.5-178 mg/L) and F (1.3-7.5 mg/L) in Cascade springs may be attributed to a magmatic source (Rose et al., 1986). The greater abundance of other major and trace components is probably derived from hydrothermal leaching and alteration of igneous rocks (Browne, 1978). Argillic alteration of amphibole and plagioclase into clays is a good source of Na (133-859 mg/L) and K (0.7-46.5 mg/L). Moderate Ca (5.4-70.9 mg/L) could come from leaching of pyroxene, amphibole, and feldspar. Trace Li (0.04-2.49) and B (0.66-9.37 mg/L) may reflect desorption from clays and other secondary minerals.

In addition to chemical enrichment, the wide range of igneous host rock lithologies may also contribute to spring chemistry diversity in the Cascades. Browne (1978) reports that fluid composition is influenced by rock type if alteration temperatures are below 280 ^{o}C. Boron, for example, may be highest in Baker (2.65 mg/L) and Ohanapecosh (9.37 mg/L) springs because of regional dominance of andesitic rocks. Silica is lowest among all thermal waters in Carson (62.3 mg/L) and Bonneville (58.6 mg/L) since they are hosted by basalt, which has low SiO_2 content. These relationships between volcanic rocks and hot waters were established by early leaching experiments by Ellis and Mahon (1964, 1966, and 1968).

In general, springs in the Olympic Mountains are more dilute and less chemically diverse than those in the Cascades. The origins of Na (63.3-75.2 mg/L), Cl (6.6-12.6 mg/L), F (0.8-1.7 mg/L), and B (0.78-1.20 mg/L) may be linked to the host rocks. The turbidite facies of the Olympics were once submarine before they had been accreted. Resulting connate waters or salts may be partially responsible for the moderate salinity of the Olympic springs. Other rock-water interactions could also be controlling spring chemistry. Low-grade alteration of feldspars would also supply considerable sodium. Leaching of interbedded carbonates and shales may influence HCO_3 (124-169 mg/L) and SO_4 (13.3-20.8 mg/L) content.

Given the available data, these rock-water relationships are good preliminary diagnostic tools. However, other factors (biogenic, atmospheric, etc.) should be considered, as they may have some significant impact on solute input, too.

6.2 Geothermal System Features

Results of isotopic analysis, geothermometry, and chemical comparisons provide constraints on the architecture of geothermal systems that underlie these thermal springs. The Cascade and Olympic thermal springs appear to represent three types of geothermal environments (Figure 6) whose features are discussed below:

6.2.1 Source water

The O-H isotopic signatures of both Cascade and Olympic springs lie on or very close to the GWML (Figure 4). Hence, their geothermal systems appear to be recharged with meteoric waters.

6.2.2 Heat source

The Cascade Range is characterized by widespread volcanic activity, so deep magmatic intrusions are likely providing heat to the thermal springs (Figures 6a and 6b). More locally, Czajkowski et al. (2014) also report that intersecting fault zones control upflow in Carson and Bonneville springs. Geothermal fluids in the Cascades may also be hotter than those in the Olympics due to magmatic fluid contribution, as indicated by elevated chloride concentrations.

In the Olympics, bounding, subvertical thrust faults probably allow meteoric waters to travel to greater depths and become heated by the geothermal gradient (Figure 6c). This hypothesis would suggest that Olympic springs are manifestations of one geothermal system or very similar multiple systems, which may explain the chemical homogeneity seen among springs in the Olympic complex and Sol Duc. In fact, all the accessible and known thermal springs in the Olympic Mountains are featured in this study and occur in close proximity to one another.

6.2.3 Reservoir connectivity

Geothermometry results show that there is a greater temperature difference between discharge and the subsurface in the Cascade springs, suggesting these fluids have taken a more indirect and/or slower path to the surface. A more circuitous route could also explain the greater role for mixing and degassing in Cascade waters.

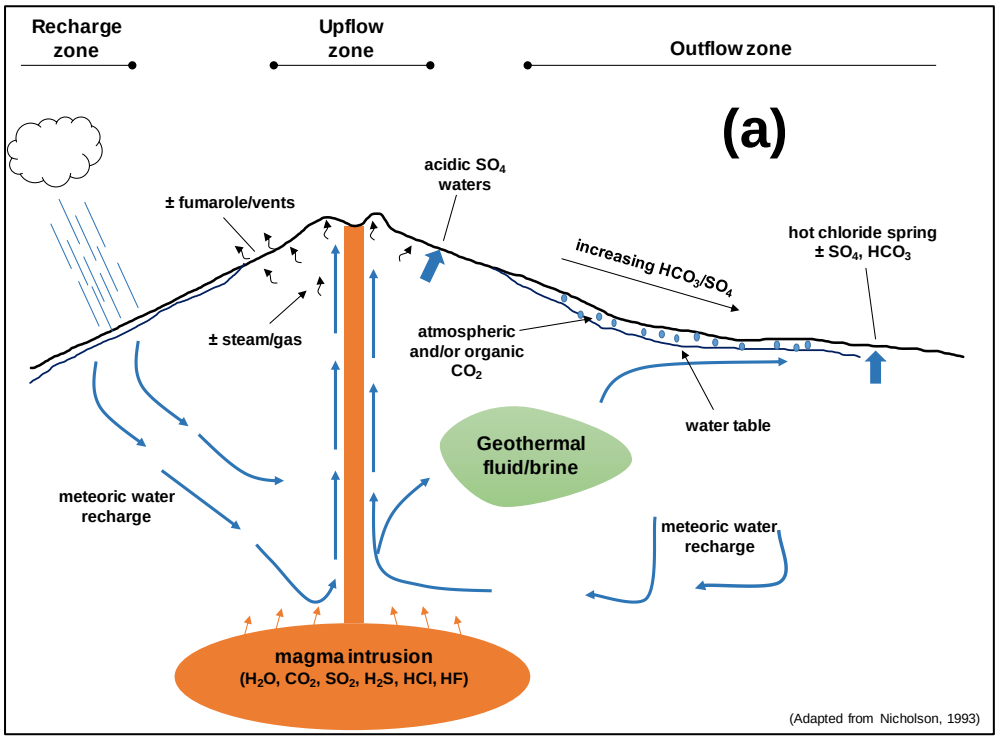
Conversely, Olympic springs appear to be more directly connected to their reservoirs, so there is a smaller difference in discharge and estimated reservoir temperatures. This notion is further supported by MME results, as all five Olympic springs are apparently still in equilibrium with their last reservoir.

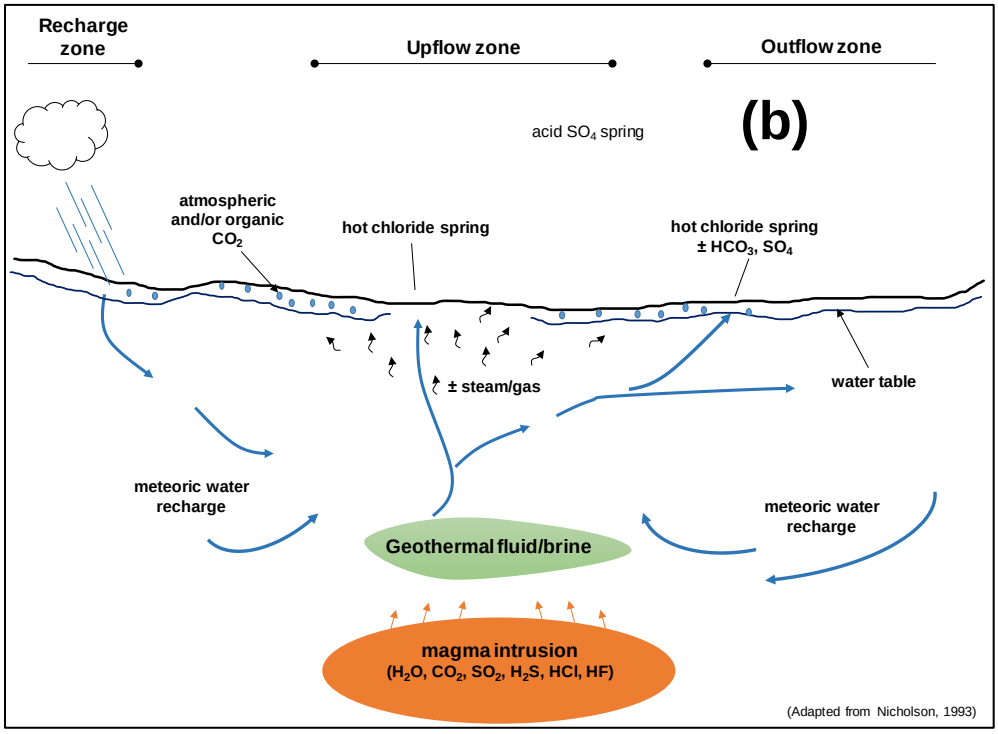
6.2.4 Topography

Differences in major anion dominance among the thermal springs can be explained by topography (Figures 6a and 6b). Ohanapecosh, Baker, and Goldmyer springs occur at high elevation where greater local relief promotes lateral flow of juvenile, acidic (SO4-rich) and shallow groundwater (HCO3-rich) (Figure 6a; Nicholson, 1993). This allows these waters to retain intermediate chemical signatures (Figure 3b). Ohanapecosh is confirmed as an outflow-type spring by the presence of travertine deposits (Frank, 1994). Similarly, the addition and dominance of HCO_3 to the chemically dilute Olympic Complex and Sol Duc spring waters may be attributed to mixing with laterally flowing groundwater reacting with pressurized atmospheric and soil CO_2 (Figures 3b and 6c).

On the other hand, Carson and Bonneville springs are seated in a low-relief environment where thermal waters are less susceptible to mixing with acidic and bicarbonate waters (Figure 6b). Even if a spring like Carson is prone to shallow groundwater dilution, it still retains a Cl-rich composition typical of a mature geothermal fluid (Figure 3b; Ellis and Mahon, 1977). However,

the absence of relief does not entirely preclude mixing with laterally-flowing waters, as is the case with Bonneville spring's slightly intermediate composition (Figures 3b and 6b).





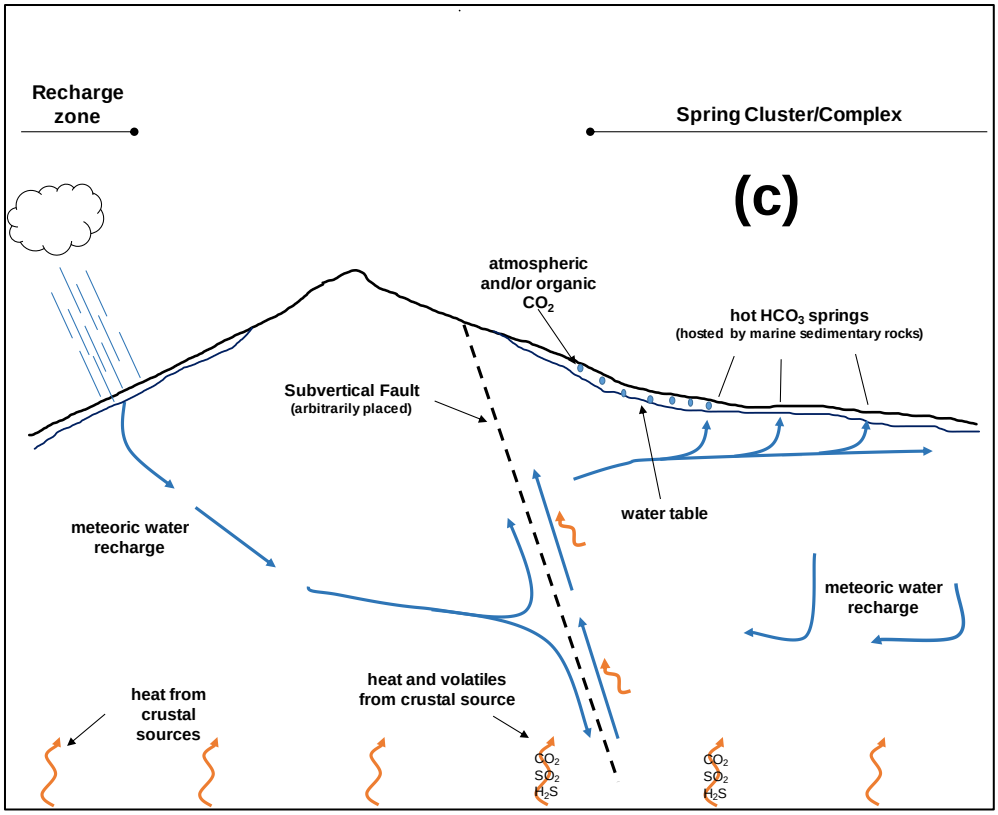


Figure 6: General conceptual models of potential Cascade and Olympic geothermal systems. Featured are: a) high-relief magmatic system, b) low-relief magmatic system, and c) fault-controlled sedimentary system.

7. Conclusions

Geochemical characterization is a powerful tool for determining the origins of geothermal fluids. The Cascade Range thermal springs are significantly more saline and diverse due to: (1) greater extent of rock-water interactions (leaching and low-grade alteration) with various igneous bedrock, (2) influences of magmatic fluids, and (3) the existence of greater topographic relief. Geothermal fluids in this province are also inherently hotter but appear to follow a more indirect path to the surface, as supported by evidence of chemical evolution (dilution and degassing) and greater differences between spring discharge and reservoir temperatures. On the other hand, the thermal springs in the Olympic Mountains are more chemically dilute, more homogeneous, and derived from a cooler subsurface reservoir. Waters in this region are moderately salinized by submarine turbidites and, to some extent, rock-water interactions as well. Small differences in surface and subsurface temperatures and undisturbed multiple mineral equilibria suggest Olympic waters have a more direct fluid network. These characteristics are largely attributed to subvertical conduits (reoriented thrust faults) that promote fluid transport. Concordance of thermal water isotopic signatures to meteoric water lines indicate that both geothermal provinces are recharged by meteoric sources.

Coupling these results with other resource characterization techniques could yield additional insights into the geothermal systems in western Washington. In particular, a better

understanding of structural controls, subsurface geology and thermal regime, along with gas sampling could improve the conceptual models and geochemical interpretations.

8. Acknowledgements

Funding for this research was provided by a UPS Agricola Scholar undergraduate research fellowship and a 2016 GRC Undergraduate scholarship. We thank fellow UPS summer researchers, Andrew Oberhelman, Mitchell Dodo, and Angie Calderon, for field assistance, Marfa Scheratski and the Northwest Wilderness Programs for allowing us sample on their property, and Mark Reed and Jim Palandri for SOLVEQ. Golla also expresses gratitude to his dad, Glenn, to Christovik Simatupang, and to Ken Clark for all their advice.

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